

JIAHAO TIAN

Hangzhou, Zhejiang, China

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Education

School of Mathematical Sciences, Zhejiang University

Aug. 2023 – Jun. 2027 (Expected)

Bachelor of Science in Statistics

Hangzhou, Zhejiang, China

- **GPA:** 88.31/100 or 3.95/4.30
- **Honors:** Selected for Chu Kochen Honors College (Physics); transferred to Statistics for strict mathematical training.

Selected Coursework

- SDEs (Grad)
- Asymptotic Statistics (Grad)
- Mathematical Statistics (Score: 95)
- Functional Analysis & Topology
- Stochastic Processes (Score: 99)
- Probability Theory (Score: 92)

Research Experience

Robustness of Knockoff Aggregation in Extreme Settings

Sept. 2025 – Present

Research Assistant | Advised by Prof. Lijun (ZJU)

Hangzhou, China

- Investigated the finite-sample FDR control of e-value-based derandomized knockoffs, identifying a severe power breakdown (Power $\rightarrow 0$) under heavy-tailed noise (Cauchy) and high feature collinearity ($\rho \geq 0.5$).
- Diagnosed the failure mechanism within the e-BH thresholding step and explored the robustness of other aggregation methods under heavy-tailed distribution; systematically evaluated the robustness of alternative statistics (e.g., LASSO path difference, Random Forest).
- Proposed a theoretical approximation framework utilizing the Glivenko-Cantelli theorem to calibrate the α_{kn}/α_{ebh} ratio and restore selection power. **First-author manuscript in preparation.**

Uncertainty Quantification via Conformal Prediction

Mar. 2026 – Present

Research Assistant | Advised by Prof. Guanhua (UW-Madison)

Remote

- Investigating conformal inference frameworks to construct distribution-free predictive intervals for complex statistical models.
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Academic Activities & Presentations

Talk: Convergence of Random Series and Large Deviations

Fall 2025

Advanced Probability Theory Seminar, Zhejiang University

Hangzhou, China

- Presented a 90-minute lecture on advanced limit theorems and large deviation principles to peers and faculty, demonstrating mastery of measure-theoretic probability.

Talk: Robustness of Multiple Knockoff Methods

Fall 2025

Weekly Statistical Seminar, Advised by Prof. Lijun Wang

Hangzhou, China

- Delivered a comprehensive review of Model-X and derandomized knockoff procedures, discussing variable selection stability and finite-sample FDR control mechanisms.

Selected Course Projects

C-Based CNN Forward Inference Engine | C, Model Optimization, Binary I/O

Spring 2025

- Implemented a C-based CNN forward inference engine from scratch, achieving a 75% latency reduction via channel pruning while maintaining $\sim 99\%$ accuracy.

Uncertainty Quantification for PDEs in Random Domains | Python, Stochastic Galerkin

Fall 2025

- Reproduced numerical solutions for elliptic SPDEs in random domains utilizing stochastic mappings, Karhunen-Loève expansions, and Legendre polynomial chaos.

Teaching & Academic Competitions

Teaching Assistant | Probability and Mathematical Statistics

Sept. 2026 – Expected

School of Mathematical Sciences, Zhejiang University

Hangzhou, China

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Shing-Tung Yau College Student Mathematics Contest

May. 2026

Contestant | Probability & Statistics Track

Hangzhou, China

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Advanced Reading & Independent Study

Measure-Theoretic Probability | *Probability: Theory and Examples (R. Durrett)*

Fall 2025

- Focused on the rigorous foundations of probability, mastering topics including the Laws of Large Numbers, Central Limit Theorems, conditional expectation, and discrete-time martingales.

Statistical Machine Learning | *The Elements of Statistical Learning (Hastie et al.)*

Spring 2025

- Conducted targeted readings including splines, Random Forests, leveraging the ESL-CN framework (initiated by advisor) to build deep intuition for high-dimensional statistical models.

Skills & Tests

Programming: Python, C/C++, R, MATLAB, Shell

Tools & OS: Linux, Git/GitHub, LaTeX, VS Code

Standardized Tests: GRE: 313 (Q170, V143, W3.5) | TOEFL: 5.5/6.0(Equal to Old Scale 110/120)